

1. Input

Section, materials and resistance basis exactly as analysed.

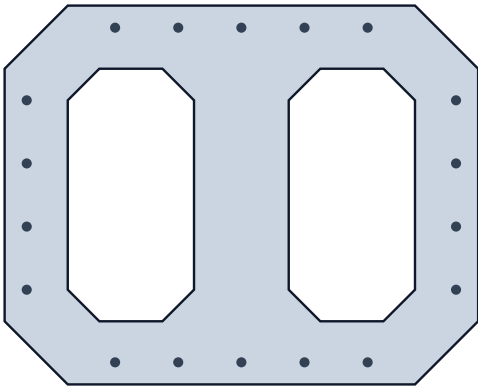
Identity

Project	PM-advanced (7) 2D reference case	Section	PM-advanced (7) 2D — 1500x1200 .
Calculation profile	KDS 2024 — Stress-strain integratio.	Mechanics	Stress-strain integration
Resistance profile	KDS 2024 current set · resistance: K.	Profile edition	2024 partial-revision framework (effe.
Verification status	draft	Concrete model basis	Code-default
Method id	kds-2024-current-set-global-strength.	Units	mm · N · MPa; forces reported in kN,.
Sign convention	Compression positive; Mx = ΣF·(y-yc.		

Section

Cross-section (mm, about the net concrete centroid)

18 bars



Section properties

Gross concrete area Ag	1,120,000 mm²
Reinforcement area As	14,476 mm²
Reinforcement ratio ρ	1.293 %
Bar count	18
Net centroid	(0.0, 0.0) mm
Bounding box	1,500.0 × 1,200.0 mm
Solids / holes	1 / 2
Perimeter	9,262.7 mm

Materials

Name	KDS C30 (workbook Input)	Standard	KDS
fck	30.0 MPa	Model	kds-parabolic
εCU	0.00330	εc0	0.00200
α	0.8500	Ec	28,417 MPa
Tension	ignored	Name	SD400 (workbook Input)
Standard	KDS	fy	400.0 MPa
Es	200,000 MPa	Model	elastic-perfectly-plastic
ey	0.002000	εU	not declared
Grades used	1		

Design resistance basis

Format	Global resultant strength-reduction f.	Transverse reinforcement	other
φ compression (ties)	0.650	φ compression (spiral)	0.700
φ tension	0.850	et,limit	0.00500
Transition rule	fixed-or-yield-multiple	Axial cap	0.800 × factored compression pole

Sampling and numerical evidence

Surface points	2024	Directions	92
Stations	22	Direction tolerance	0.500 %
Worst interpolation error	0.4792 %	Tolerance reached	yes
Concrete discretization	864 cells / 5196 points	Strain domain	concrete-pivot-ultimate

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is integrated from the material stress-strain law over a verified triangle/quadrature mesh of the section.

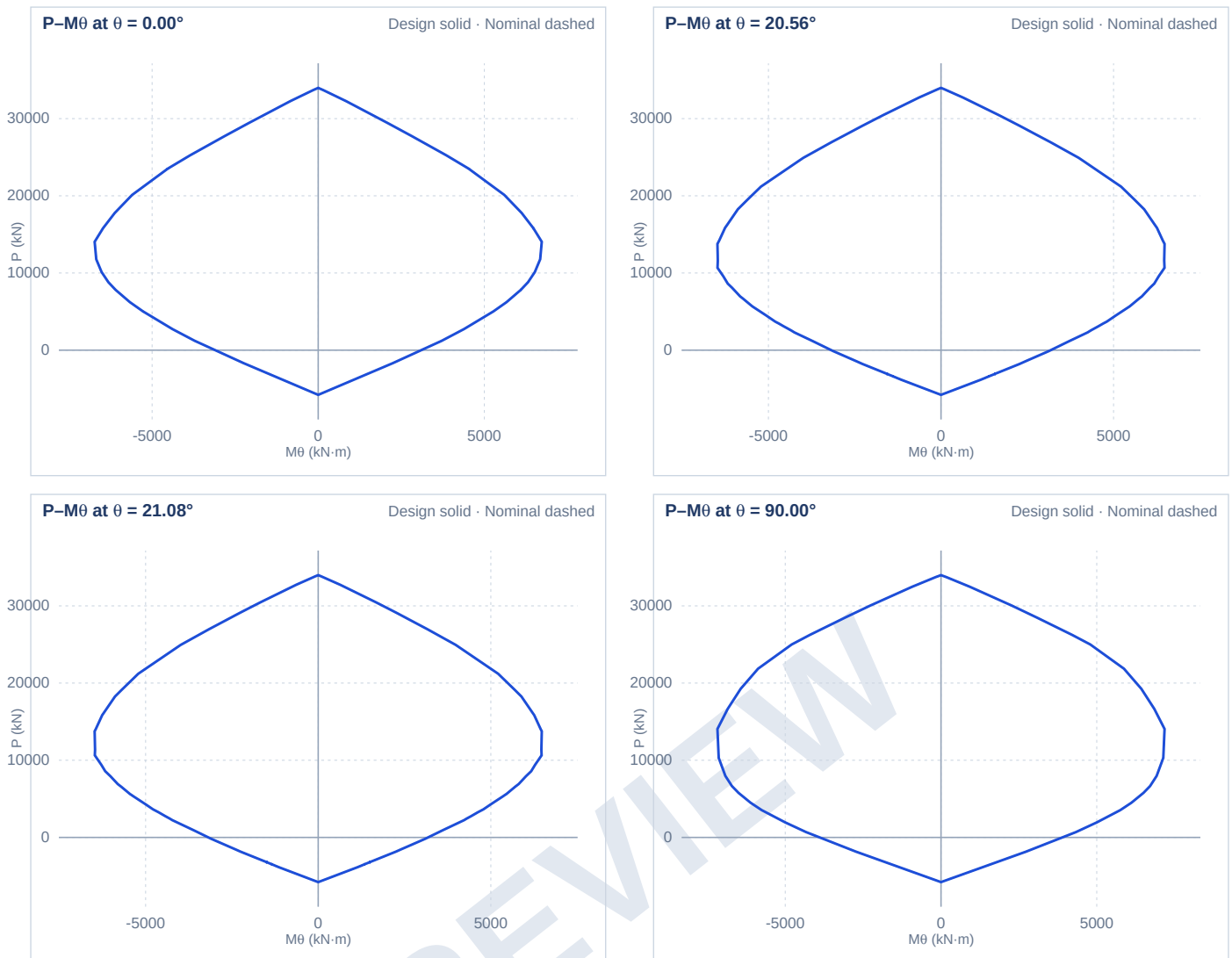
Reinforcement schedule

#	Bar id	(mm)	X (mm)	Y (mm)	As (mm ²)	Steel
1	1	32.0	-400.0	530.0	804.2	SD400 (workbook Input)
2	2	32.0	-200.0	530.0	804.2	SD400 (workbook Input)
3	3	32.0	0.0	530.0	804.2	SD400 (workbook Input)
4	4	32.0	200.0	530.0	804.2	SD400 (workbook Input)
5	5	32.0	400.0	530.0	804.2	SD400 (workbook Input)
6	6	32.0	-400.0	-530.0	804.2	SD400 (workbook Input)
7	7	32.0	-200.0	-530.0	804.2	SD400 (workbook Input)
8	8	32.0	0.0	-530.0	804.2	SD400 (workbook Input)
9	9	32.0	200.0	-530.0	804.2	SD400 (workbook Input)
10	10	32.0	400.0	-530.0	804.2	SD400 (workbook Input)
11	11	32.0	-680.0	300.0	804.2	SD400 (workbook Input)
12	12	32.0	-680.0	100.0	804.2	SD400 (workbook Input)
13	13	32.0	-680.0	-100.0	804.2	SD400 (workbook Input)
14	14	32.0	-680.0	-300.0	804.2	SD400 (workbook Input)
15	15	32.0	680.0	300.0	804.2	SD400 (workbook Input)
16	16	32.0	680.0	100.0	804.2	SD400 (workbook Input)
17	17	32.0	680.0	-100.0	804.2	SD400 (workbook Input)
18	18	32.0	680.0	-300.0	804.2	SD400 (workbook Input)

PREVIEW

2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.



3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	Pu (kN)	Mux (kN·m)	Muy (kN·m)	θ_L (°)	ϕ	Class	UR	Verdict
1	ULS-1 workbook Newton d.	24942.9	3714.2	1431.8	21.1	—	—	1.000	ADEQUATE
2	ULS-2 same axial, half mo.	24942.9	1857.1	715.9	21.1	—	—	0.860	ADEQUATE
3	ULS-3 low axial, high mom.	5000.0	4000.0	1500.0	20.6	—	—	0.743	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

PREVIEW

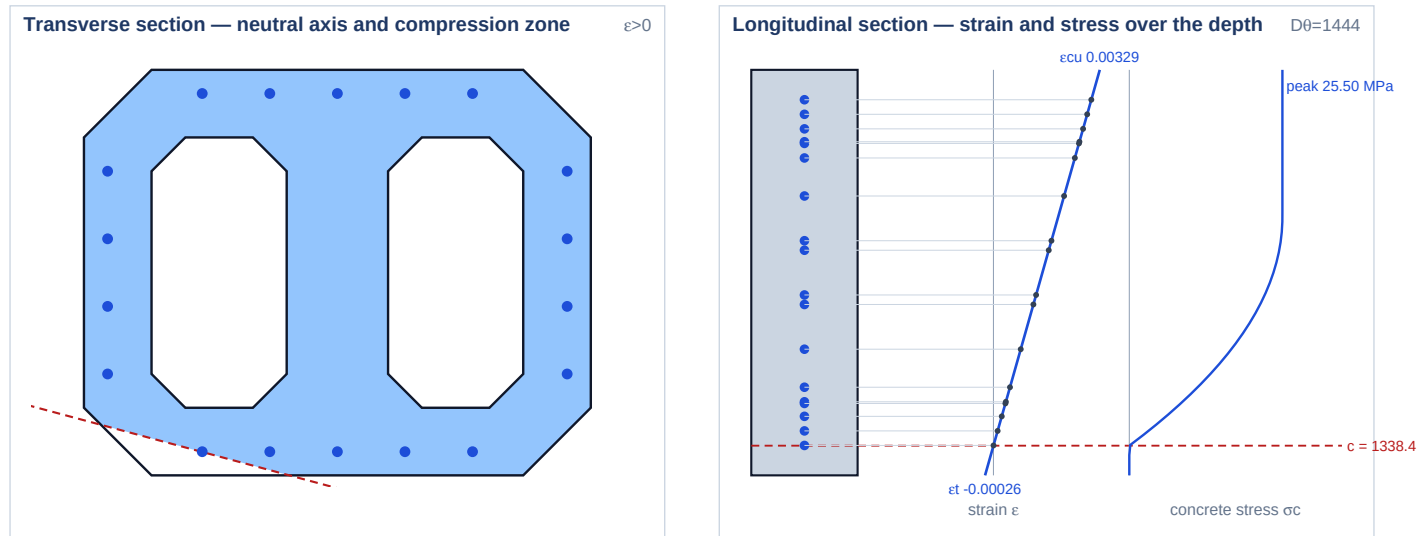
4.1 ULS-1 workbook Newton demand (on the 15 deg surface)

$P_u = 24942.9 \text{ kN}$ · $M_{ux} = 3714.2 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1431.8 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

ADEQUATE · utilization 0.9998 · fixed-P diagnostic 0.9993

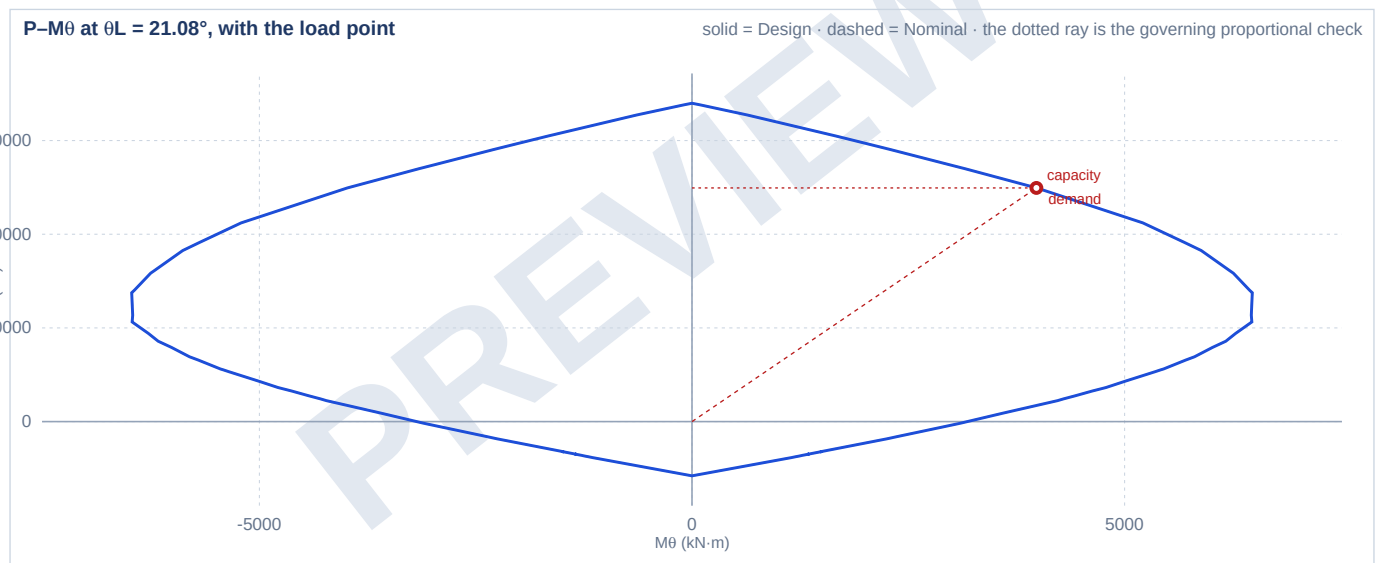
Converged preview equilibrium inside the material domain.

Section state



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Interaction diagram through the demand direction



Concrete and resultants

Concrete force	21,577.4 kN	Concrete Mx	2,900.4 kN·m
Concrete My	1,122.3 kN·m	Steel force (gross)	3,629.5 kN
Displaced concrete	-264.0 kN	Steel net	3,365.5 kN
Response P	24,942.9 kN	Response Mx	3,714.2 kN·m
Response My	1,431.8 kN·m	Residual $ \Delta P $	1.01e-4 N
Residual $ \Delta M_x $	3.15e-3 N·mm	Residual $ \Delta M_y $	4.99e-3 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ε	σ_s (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.8	2.520e-3	400.00	321.70	—
2	-200.0	530.0	804.2	262.0	2.648e-3	400.00	321.70	—
3	0.0	530.0	804.2	210.1	2.775e-3	400.00	321.70	—
4	200.0	530.0	804.2	158.3	2.903e-3	400.00	321.70	—

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
5	400.0	530.0	804.2	106.5	3.030e-3	400.00	321.70	—
6	-400.0	-530.0	804.2	1337.6	1.971e-6	0.39	0.32	—
7	-200.0	-530.0	804.2	1285.8	1.295e-4	25.89	20.82	—
8	0.0	-530.0	804.2	1233.9	2.570e-4	51.39	41.33	—
9	200.0	-530.0	804.2	1182.1	3.845e-4	76.89	61.84	—
10	400.0	-530.0	804.2	1130.3	5.120e-4	102.39	82.35	—
11	-680.0	300.0	804.2	608.5	1.795e-3	359.09	288.80	—
12	-680.0	100.0	804.2	801.7	1.320e-3	264.06	212.37	—
13	-680.0	-100.0	804.2	994.8	8.451e-4	169.02	135.94	—
14	-680.0	-300.0	804.2	1188.0	3.699e-4	73.99	59.50	—
15	680.0	300.0	804.2	256.1	2.662e-3	400.00	321.70	—
16	680.0	100.0	804.2	449.2	2.187e-3	400.00	321.70	—
17	680.0	-100.0	804.2	642.4	1.712e-3	342.42	275.39	—
18	680.0	-300.0	804.2	835.6	1.237e-3	247.38	198.96	—

Solver and admissibility evidence

Strain plane ϵ_0	1.5162e-3	Curvature k_x	2.3759e-6
Curvature k_y	6.3748e-7	Neutral-axis line	164.9806°
Compression depth c	1,338.387 mm	Iterations	7
Relative residual	2.576e-12	Max concrete compression	0.003292
Concrete limit	0.003300	Max steel tension	0.000000
Steel limit	not declared	Strain admissibility	admissible
Governing utilization	0.9998		

4.2 ULS-2 same axial, half moment (inside)

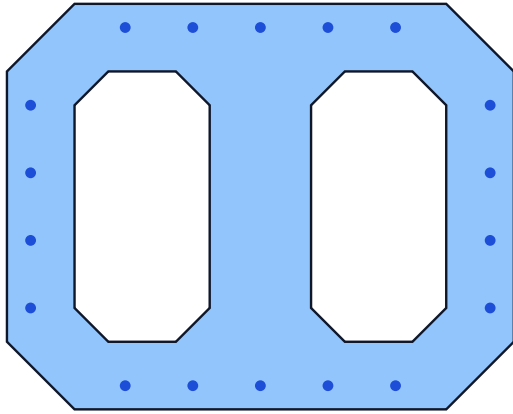
$P_u = 24942.9 \text{ kN}$ · $M_{ux} = 1857.1 \text{ kN}\cdot\text{m}$ · $M_{uy} = 715.9 \text{ kN}\cdot\text{m}$ · $\theta_L = 21.08^\circ$

ADEQUATE · utilization 0.8603 · fixed-P diagnostic 0.4996

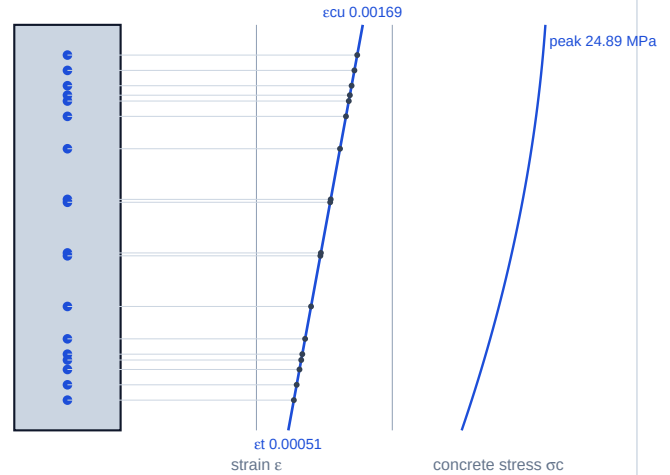
Converged preview equilibrium inside the material domain.

Section state

Transverse section — neutral axis and compression zone $\varepsilon > 0$



Longitudinal section — strain and stress over the depth $D\theta = 1456$

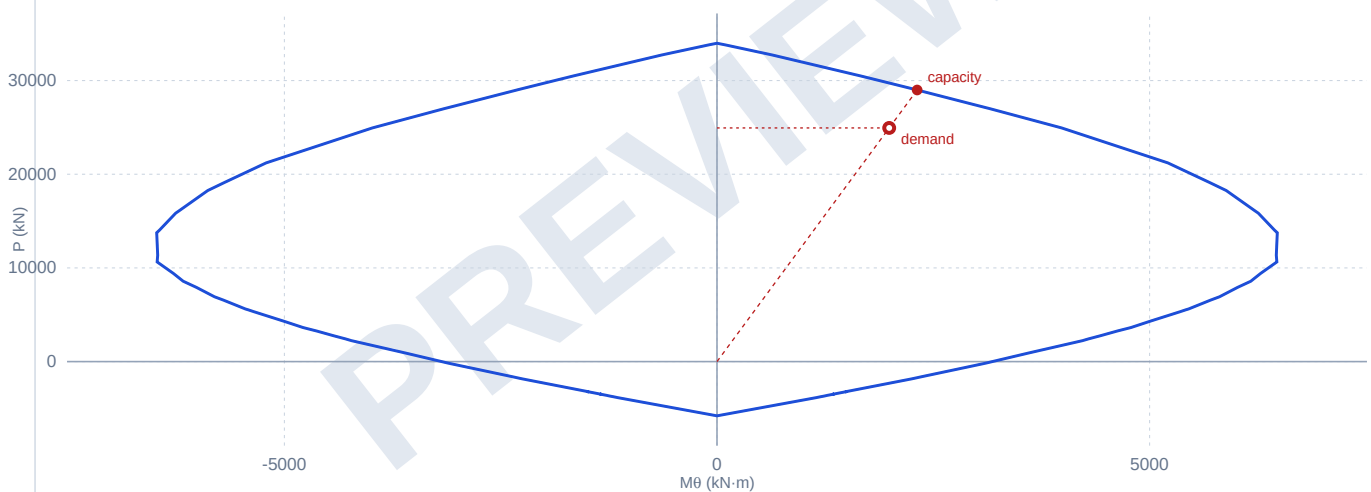


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Interaction diagram through the demand direction

P-M θ at $\theta_L = 21.08^\circ$, with the load point

solid = Design · dashed = Nominal · the dotted ray is the governing proportional check $c = 2079.0$



Concrete and resultants

Concrete force **22,045.7 kN**
 Concrete My **563.3 kN·m**
 Displaced concrete **-282.9 kN**
 Response P **24,942.9 kN**
 Response My **715.9 kN·m**
 Residual $|\Delta M_x|$ **5.19e-3 N·mm**

Concrete Mx **1,476.8 kN·m**
 Steel force (gross) **3,180.1 kN**
 Steel net **2,897.2 kN**
 Response Mx **1,857.1 kN·m**
 Residual $|\Delta P|$ **1.43e-5 N**
 Residual $|\Delta M_y|$ **1.92e-3 N·mm**

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ε	σ_s (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	328.6	1.423e-3	284.64	228.92	—
2	-200.0	530.0	804.2	273.6	1.468e-3	293.59	236.12	—
3	0.0	530.0	804.2	218.6	1.513e-3	302.54	243.32	—
4	200.0	530.0	804.2	163.6	1.557e-3	311.49	250.51	—

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
5	400.0	530.0	804.2	108.6	1.602e-3	320.43	257.71	—
6	-400.0	-530.0	804.2	1347.7	5.946e-4	118.92	95.64	—
7	-200.0	-530.0	804.2	1292.7	6.393e-4	127.87	102.84	—
8	0.0	-530.0	804.2	1237.7	6.841e-4	136.81	110.03	—
9	200.0	-530.0	804.2	1182.7	7.288e-4	145.76	117.23	—
10	400.0	-530.0	804.2	1127.7	7.735e-4	154.71	124.42	—
11	-680.0	300.0	804.2	626.8	1.181e-3	236.16	189.93	—
12	-680.0	100.0	804.2	819.1	1.024e-3	204.89	164.78	—
13	-680.0	-100.0	804.2	1011.4	8.681e-4	173.62	139.63	—
14	-680.0	-300.0	804.2	1203.7	7.118e-4	142.35	114.49	—
15	680.0	300.0	804.2	252.7	1.485e-3	297.00	238.86	—
16	680.0	100.0	804.2	444.9	1.329e-3	265.73	213.71	—
17	680.0	-100.0	804.2	637.2	1.172e-3	234.46	188.57	—
18	680.0	-300.0	804.2	829.5	1.016e-3	203.19	163.42	—

Solver and admissibility evidence

Strain plane ϵ_0	1.0984e-3	Curvature k_x	7.8173e-7
Curvature k_y	2.2369e-7	Neutral-axis line	164.0318°
Compression depth c	2,079.010 mm	Iterations	5
Relative residual	3.635e-13	Max concrete compression	0.001690
Concrete limit	0.003300	Max steel tension	0.000000
Steel limit	not declared	Strain admissibility	admissible
Governing utilization	0.8603		

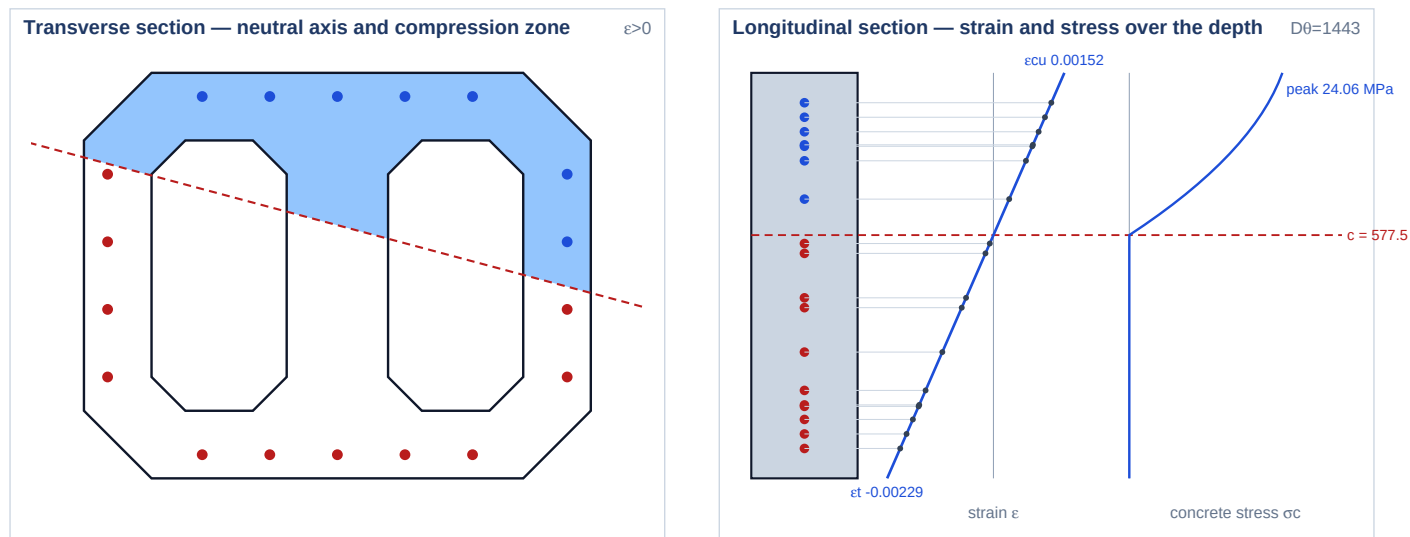
4.3 ULS-3 low axial, high moment

$P_u = 5000.0 \text{ kN}$ · $M_{ux} = 4000.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 1500.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 20.56^\circ$

ADEQUATE · utilization 0.7425 · fixed-P diagnostic 0.8156

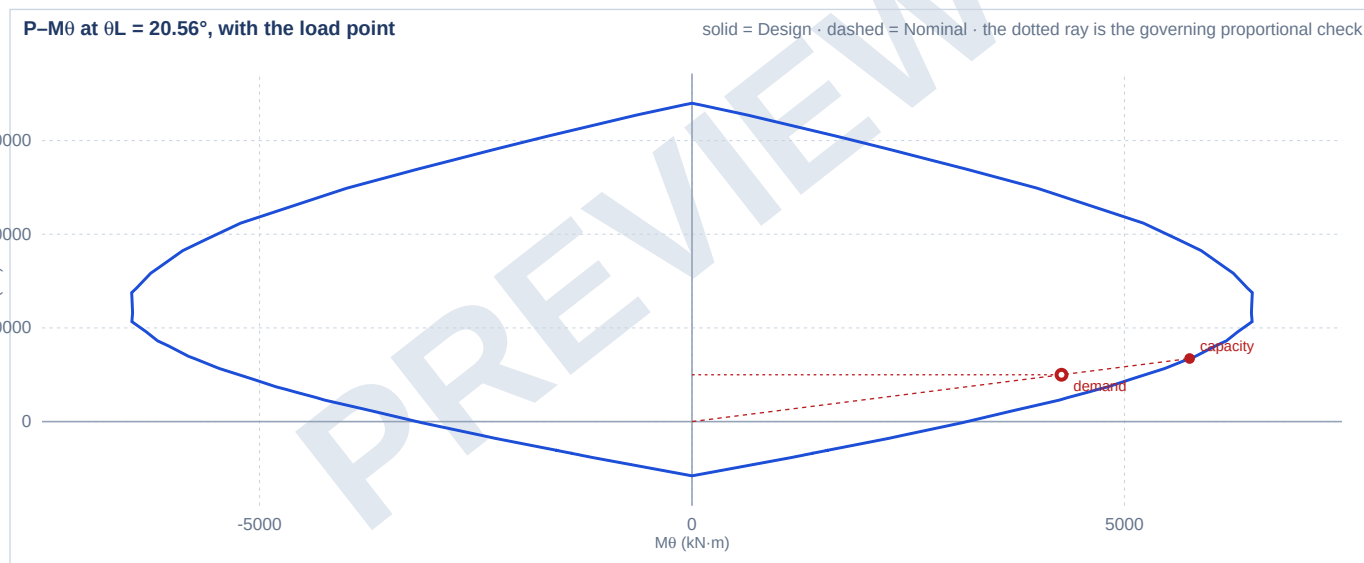
Converged preview equilibrium inside the material domain.

Section state



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Interaction diagram through the demand direction



Concrete and resultants

Concrete force	6,195.6 kN	Concrete Mx	2,728.2 kN·m
Concrete My	1,023.3 kN·m	Steel force (gross)	-1,101.1 kN
Displaced concrete	-94.5 kN	Steel net	-1,195.6 kN
Response P	5,000.0 kN	Response Mx	4,000.0 kN·m
Response My	1,500.0 kN·m	Residual $ \Delta P $	5.59e-9 N
Residual $ \Delta Mx $	0.00e+0 N·mm	Residual $ \Delta My $	9.54e-7 N·mm

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ε	σ_s (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.0	6.983e-4	139.66	112.33	—
2	-200.0	530.0	804.2	261.4	8.347e-4	166.94	134.26	—
3	0.0	530.0	804.2	209.7	9.711e-4	194.22	156.20	—
4	200.0	530.0	804.2	158.0	1.108e-3	221.50	178.14	—

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
5	400.0	530.0	804.2	106.4	1.244e-3	248.78	200.08	—
6	-400.0	-530.0	804.2	1337.1	-2.005e-3	-400.00	-321.70	—
7	-200.0	-530.0	804.2	1285.4	-1.869e-3	-373.74	-300.58	—
8	0.0	-530.0	804.2	1233.7	-1.732e-3	-346.46	-278.64	—
9	200.0	-530.0	804.2	1182.1	-1.596e-3	-319.18	-256.70	—
10	400.0	-530.0	804.2	1130.4	-1.460e-3	-291.91	-234.76	—
11	-680.0	300.0	804.2	607.6	-7.922e-5	-15.84	-12.74	—
12	-680.0	100.0	804.2	800.8	-5.893e-4	-117.86	-94.79	—
13	-680.0	-100.0	804.2	994.0	-1.099e-3	-219.88	-176.84	—
14	-680.0	-300.0	804.2	1187.2	-1.609e-3	-321.89	-258.88	—
15	680.0	300.0	804.2	256.2	8.483e-4	169.65	136.44	—
16	680.0	100.0	804.2	449.4	3.382e-4	67.64	54.40	—
17	680.0	-100.0	804.2	642.7	-1.719e-4	-34.38	-27.65	—
18	680.0	-300.0	804.2	835.9	-6.820e-4	-136.40	-109.70	—

Solver and admissibility evidence

Strain plane ϵ_0	-3.8060e-4	Curvature k_x	2.5504e-6
Curvature k_y	6.8197e-7	Neutral-axis line	165.0295°
Compression depth c	577.545 mm	Iterations	6
Relative residual	1.419e-16	Max concrete compression	0.001525
Concrete limit	0.003300	Max steel tension	0.002005
Steel limit	not declared	Strain admissibility	admissible
Governing utilization	0.7425		